

Stochastic problems

What is Stochastics

- ◆ Randomness based problems
- ◆ Analyze using statistics
 - ◆ Hope your *statistically* correct

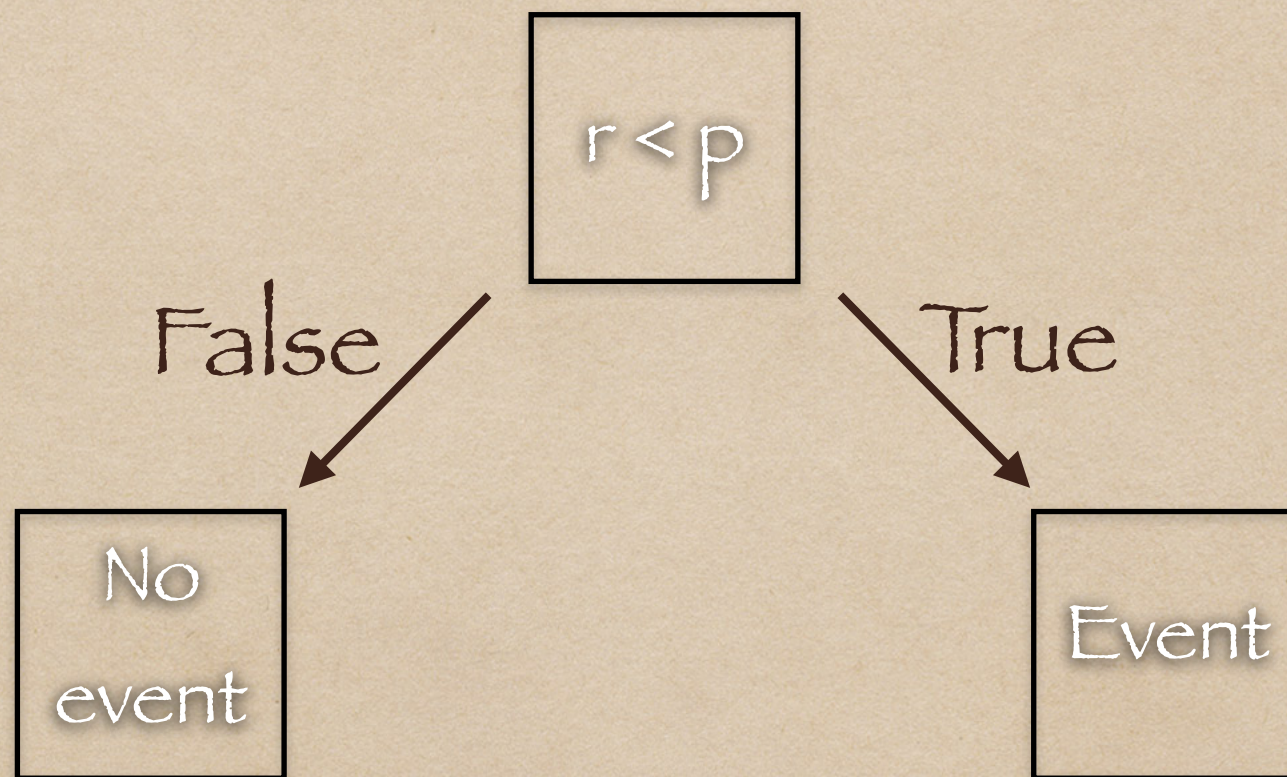
Generate problems

- ◆ Using random numbers $\rightarrow randi$
- ◆ Using rationals i.e. $\frac{\mathbb{Z}}{\mathbb{Z}}$ for prob

Simulation

r = random number

p = probability



MATLAB

```
p = prob. level;           % the probability of event in decimal form
r = rand;                  % generate a random number from 0 to 1
if r < p                    % this should happen about p % of the time
    event happens
else                        % if event does not happen, alternate event
    alternate event
end
```


Example

```
p = 0.75;           % 75% chance of reproducing
Pop = 2000;         % current population of bacteria
Live = 0;           % initialize counters
Die = 0;
for BactNum = 1:Pop % for each bacterium in pop.
    r = rand;        % generate a random number
    if r < p          % determine if it divides or dies
        Live = Live + 1 % count divisions
    else
        Die = Die + 1; % count deaths
    end
end
```


Problems

- ◆ Independent vs dependent events
- ◆ Graphing p vs the reality

Common Error

- ◆ Error from discrete space
- ◆ Error induced by random generation